

Signature (4,4) Einstein-Spinor Field Equations in Component Form on Coordinates $\{\mathbf{x}_0, \dots, \mathbf{x}_7\}$

Complete coordinate-explicit provenance report

Reproducible exact-real implementation

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Abstract

This report rewrites the curved $\text{Spin}(4,4)$ Einstein-spinor model in the exact coordinate ordering

`coordinates = {x0, x1, x2, x3, x4, x5, x6, x7}`

matching the bridge notebook convention used in `gpt-5.6_bridge.nb`. The evolution coordinate is `x4`.

The matter field is one commuting real 16-component spinor field $\mathbf{psi}=(\mathbf{psi1},\dots,\mathbf{psi16})^T$. The model includes neither a scalar quintessence field nor a cosmological constant. Every Einstein and spinor equation is presented in component form, including all vanishing off-diagonal components.

1 Geometry and index conventions

Tangent-signature matrix:

$$\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

Coordinate indices:

$$\mu, \nu, \rho, \sigma \in \{0, 1, 2, 3, 4, 5, 6, 7\}, \quad I, J \in \{0, 1, 2, 3, 5, 6, 7\}.$$

Metric ansatz:

$$g_{\mu\nu} = \text{diag}(a(x4)^2, a(x4)^2, a(x4)^2, a(x4)^2, -1, a(x4)^2, a(x4)^2, a(x4)^2),$$

$$H(x4) = \frac{da/dx4}{a}.$$

2 Nonzero connection and curvature components

Levi-Civita nonzero Christoffel components:

$$\Gamma^4_{II} = \eta_{II} a \frac{da}{dx4}, \quad \Gamma^I_{4I} = \Gamma^I_{I4} = H.$$

Einstein tensor components:

$$G_{44} = 21H^2,$$

$$G_{II} = -\eta_{II} a^2 \left(6 \frac{dH}{dx4} + 21H^2 \right),$$

$$G_{\mu\nu} = 0 \quad (\mu \neq \nu).$$

3 Spinor sector and potential

Real 16-component commuting spinor field:

$$\psi(x4) = \begin{pmatrix} \psi_1(x4) \\ \psi_2(x4) \\ \vdots \\ \psi_{16}(x4) \end{pmatrix}.$$

Invariant bilinear with symmetric split matrix C :

$$S = \psi^\top C \psi.$$

Potential and derivative:

$$V(S) = \frac{1}{20}S + \frac{19}{20}S^{1/5}, \quad V_S = \frac{dV}{dS} = \frac{1}{20} + \frac{19}{100}S^{-4/5}.$$

4 Stress-energy components (all components listed)

Energy density and pressure:

$$\rho = \frac{1}{20}S + \frac{19}{20}S^{1/5}, \quad p = \left(\frac{1}{5} - 1\right) \frac{19}{20}S^{1/5} = -\frac{19}{25}S^{1/5}.$$

Component form:

$$T_{44} = \rho,$$

$$T_{II} = \eta_{II} a^2 p,$$

$$T_{\mu\nu} = 0 \quad (\mu \neq \nu).$$

5 Einstein equations in complete component form

Use

$$G_{\mu\nu} = \kappa_8 T_{\mu\nu}, \quad \kappa_8 = 21.$$

5.1 The 36 independent symmetric components

Time-time component:

$$\boxed{21H^2 = 21\rho} \iff H^2 = \rho.$$

Seven diagonal transverse components ($I \setminus \{0, 1, 2, 3, 5, 6, 7\}$):

$$\boxed{-\eta_{II} a^2 \left(6 \frac{dH}{dx^4} + 21H^2 \right) = 21\eta_{II} a^2 p}$$

which reduce to

$$\boxed{6 \frac{dH}{dx^4} + 21H^2 = -21p}.$$

Twenty-eight off-diagonal components:

$$\boxed{G_{\mu\nu} = 0 = 21 T_{\mu\nu} \text{ for every } \mu < \nu.}$$

Explicitly,

$$(\mu, \nu) \in \{(0, 1), (0, 2), (0, 3), (0, 4), (0, 5), (0, 6), (0, 7)\},$$

$$(\mu, \nu) \in \{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7)\},$$

$$(\mu, \nu) \in \{(2, 3), (2, 4), (2, 5), (2, 6), (2, 7)\},$$

$$(\mu, \nu) \in \{(3, 4), (3, 5), (3, 6), (3, 7), (4, 5), (4, 6), (4, 7)\},$$

$$(\mu, \nu) \in \{(5, 6), (5, 7), (6, 7)\}.$$

6 Spinor field equations in complete component form

Homogeneous Dirac equation:

$$\frac{d\psi}{dx^4} = -\frac{7}{2}H\psi - V_S\gamma^4\psi.$$

For each component $A=1, \dots, 16$:

$$\boxed{\frac{d\psi_A}{dx4} = -\frac{7}{2}H\psi_A - V_S \sum_{B=1}^{16} (\gamma^4)_{AB} \psi_B}.$$

Written one-by-one:

1. $\frac{d\psi_1}{dx4} = -\frac{7}{2}H\psi_1 - V_S \sum_{B=1}^{16} (\gamma^4)_{1B} \psi_B$
2. $\frac{d\psi_2}{dx4} = -\frac{7}{2}H\psi_2 - V_S \sum_{B=1}^{16} (\gamma^4)_{2B} \psi_B$
3. $\frac{d\psi_3}{dx4} = -\frac{7}{2}H\psi_3 - V_S \sum_{B=1}^{16} (\gamma^4)_{3B} \psi_B$
4. $\frac{d\psi_4}{dx4} = -\frac{7}{2}H\psi_4 - V_S \sum_{B=1}^{16} (\gamma^4)_{4B} \psi_B$
5. $\frac{d\psi_5}{dx4} = -\frac{7}{2}H\psi_5 - V_S \sum_{B=1}^{16} (\gamma^4)_{5B} \psi_B$
6. $\frac{d\psi_6}{dx4} = -\frac{7}{2}H\psi_6 - V_S \sum_{B=1}^{16} (\gamma^4)_{6B} \psi_B$
7. $\frac{d\psi_7}{dx4} = -\frac{7}{2}H\psi_7 - V_S \sum_{B=1}^{16} (\gamma^4)_{7B} \psi_B$
8. $\frac{d\psi_8}{dx4} = -\frac{7}{2}H\psi_8 - V_S \sum_{B=1}^{16} (\gamma^4)_{8B} \psi_B$
9. $\frac{d\psi_9}{dx4} = -\frac{7}{2}H\psi_9 - V_S \sum_{B=1}^{16} (\gamma^4)_{9B} \psi_B$
10. $\frac{d\psi_{10}}{dx4} = -\frac{7}{2}H\psi_{10} - V_S \sum_{B=1}^{16} (\gamma^4)_{10B} \psi_B$
11. $\frac{d\psi_{11}}{dx4} = -\frac{7}{2}H\psi_{11} - V_S \sum_{B=1}^{16} (\gamma^4)_{11B} \psi_B$
12. $\frac{d\psi_{12}}{dx4} = -\frac{7}{2}H\psi_{12} - V_S \sum_{B=1}^{16} (\gamma^4)_{12B} \psi_B$
13. $\frac{d\psi_{13}}{dx4} = -\frac{7}{2}H\psi_{13} - V_S \sum_{B=1}^{16} (\gamma^4)_{13B} \psi_B$
14. $\frac{d\psi_{14}}{dx4} = -\frac{7}{2}H\psi_{14} - V_S \sum_{B=1}^{16} (\gamma^4)_{14B} \psi_B$
15. $\frac{d\psi_{15}}{dx4} = -\frac{7}{2}H\psi_{15} - V_S \sum_{B=1}^{16} (\gamma^4)_{15B} \psi_B$
16. $\frac{d\psi_{16}}{dx4} = -\frac{7}{2}H\psi_{16} - V_S \sum_{B=1}^{16} (\gamma^4)_{16B} \psi_B$

7 Reduced autonomous ODE system solved by the study

State vector:

$$y = (a, H, \psi_1, \dots, \psi_{16}) \in \mathbb{R}^{18}.$$

Equations:

$$\frac{da}{dx^4} = aH,$$

$$\frac{dH}{dx^4} = -\frac{\kappa_8}{6}(\rho + p),$$

$$\frac{d\psi_A}{dx^4} = -\frac{7}{2}H\psi_A - V_S \sum_{B=1}^{16} (\gamma^4)_{AB} \psi_B, \quad A = 1, \dots, 16.$$

8 Model-internal dark-sector connections from the components

The component equations imply two distinguished effective pieces:

1. Dust-like term from $S/20$ with equation-of-state $w = 0$ at the term level.
2. Negative-pressure term from $(19/20)S^{1/5}$ with term-level equation-of-state $w = \alpha - 1 = -4/5$.

Because the equations are exact in this coordinate basis, these statements follow from component reduction rather than from coordinate relabeling.

9 Reproduction commands

9.1 Windows PowerShell

```
.\scripts\verify_phase7_x0_x7_reports.ps1
```

9.2 Git Bash or WSL

```
bash ./scripts/verify_phase7_x0_x7_reports.sh
```